

Description of the EG_CO2.py script's conceptual data flows

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The conceptual diagram is presented in Figure 1.

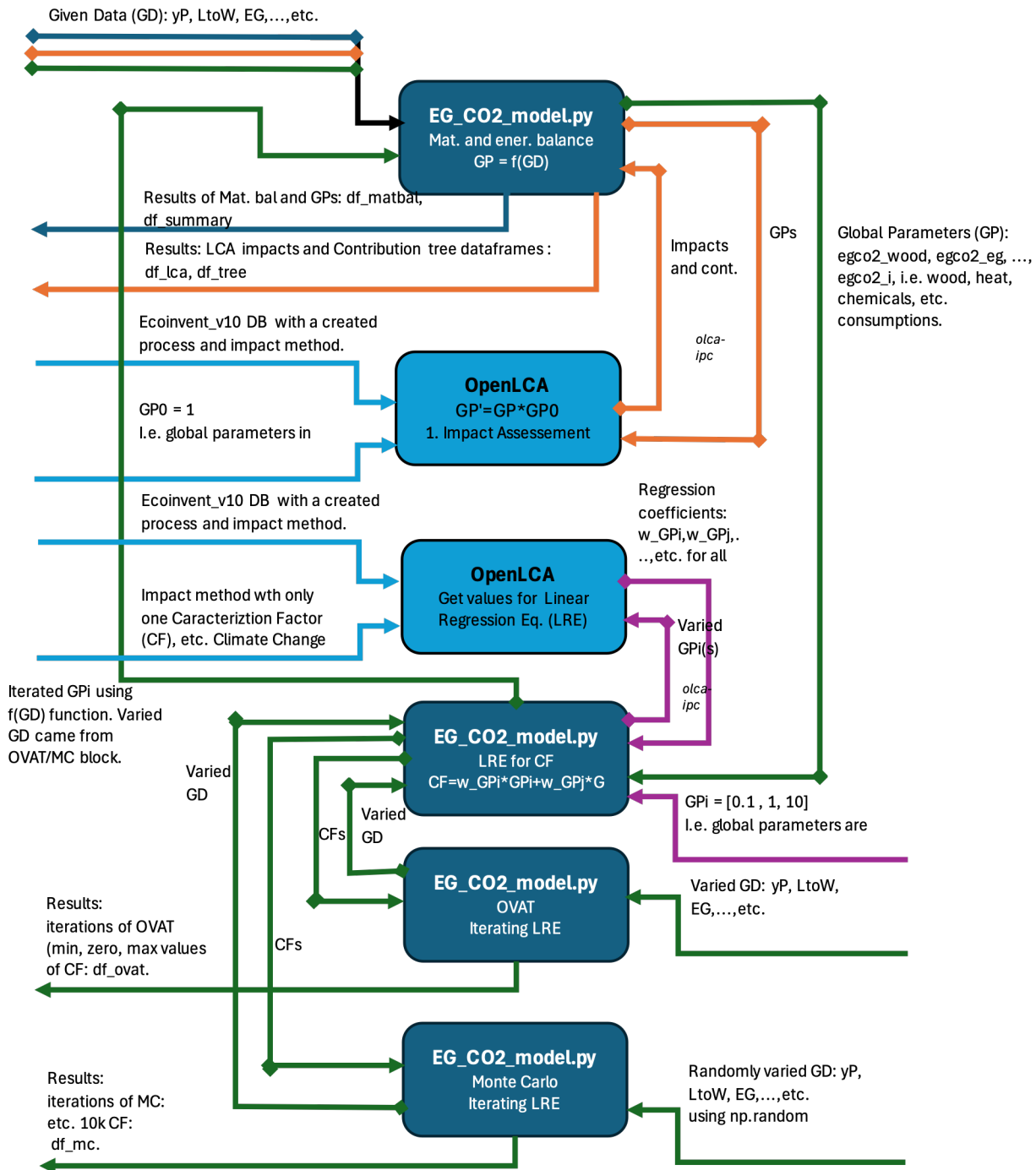


Figure 1. EG_CO2.py script's conceptual data flows diagram.

1) Inputs and core model

- Given Data (GD): user-defined process parameters such as yP, LtoW, EG, etc.
- EG_CO2_model.py computes:
 - Material and energy balance
 - Global Parameters (GP) as functions of GD: $GP = f(GD)$
 - Optional scaling: $GP' = GP \cdot GP_0$ with $GP_0 = 1$ (baseline)
- Immediate results from this block:
 - df_matbal (material/energy balance results)
 - df_summary (summary of computed GPs)

2) Life Cycle Impact Assessment (LCIA) in openLCA

- The model communicates with openLCA via olca-ipc, using an Ecoinvent_v10 database with a created process and a custom impact method (single characterization factor, e.g., climate change).
- openLCA returns:
 - df_lca (impact results)
 - df_tree (contribution tree)

3) Linear Regression Equation (LRE) for CF modeling

- Characterization Factor (CF) is modeled as a linear function of GPs:
 - $CF = w_{GPI} \cdot GPI + w_{GPj} \cdot GPj + \dots$ (regression coefficients $w_{\dots}$ are estimated)
- The LRE provides a fast surrogate to map GP changes to CFs, complementing the direct openLCA evaluations.

4) Sensitivity and uncertainty analyses

- OVAT (one-variable-at-a-time):
 - Iteratively varies a selected GPI over representative values (e.g., [0.1, 1, 10]) and/or varies GD to induce GP changes via $f(GD)$.
 - Runs the LRE and/or re-executes the openLCA assessment as needed.
 - Outputs df_ovat (CFs at min/zero/max or designated bounds).

- Monte Carlo (MC):
 - Randomly perturbs GD using np.random (e.g., 10k iterations), propagates through GP = f(GD), and evaluates CF via LRE and/or openLCA.
 - Outputs df_mc (distribution of CFs across iterations).
- Data coupling:
 - “Varied GD” originates in the OVAT/MC block and is fed back into EG_CO2_model.py to recompute GPs; those GPs are then passed to the LRE and to openLCA.

5) Data objects and their roles

- Inputs: GD (yP, LtoW, EG, ...)
- Intermediate: GP, GP'
- Results from the model: df_matbal, df_summary
- LCIA results from openLCA: df_lca, df_tree
- Sensitivity/uncertainty outputs: df_ovat (OVAT), df_mc (MC)

6) Tooling and DB

- EG_CO2_model.py for balances and GP computation; NumPy for randomization; openLCA via olca-ipc; Ecoinvent_v10 with a custom process and impact method (single CF, e.g., climate change).

High-level flow (summary)

- GD → EG_CO2_model.py → GP (and df_matbal/df_summary) → openLCA (olca-ipc, Ecoinvent_v10, custom method) → df_lca, df_tree.
- In parallel, CF is modeled via LRE using GPs ($CF = \sum w \cdot GP$).
- OVAT/MC vary GD (and implicitly GP), iterating LRE and/or openLCA:
 - OVAT → df_ovat; MC (e.g., 10k) → df_mc. Varied GD from these blocks feeds back into the model.